

2022 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Forces — Skydiving & Safety

In 2012, a stunt skydiver jumped from a helicopter at 730 m without a parachute, wearing a wingsuit, reached speeds up to 130 km h^{-1} , and landed on cardboard boxes. Comment on the physics challenges involved in carrying out the stunt successfully.

Worked discussion

- The skydiver must control descent with the wingsuit, generating lift and drag to reduce the vertical velocity and steer accurately towards the target.
- High speed ($\approx 36 \text{ m s}^{-1}$) means very large kinetic energy. Energy must be absorbed safely on impact.
- Cardboard boxes compress and crumple, increasing the stopping distance/time and reducing the average decelerating force ($F = \Delta p / \Delta t$).
- Precise targeting is essential — small errors in trajectory could miss the box stack.
- Air resistance is crucial: it reduces the terminal speed compared with free fall, but still leaves a dangerous kinetic energy to dissipate.
- Physiological challenge: the skydiver must withstand

rapid deceleration forces while remaining conscious and uninjured.

Final points

Challenges: controlling flight, managing large kinetic energy, dissipating energy safely through crumpling boxes, and ensuring precise landing on target.

Revision tips

- Kinetic energy $KE = \frac{1}{2}mv^2$: very large at 130 km h⁻¹.
- Safety requires increasing stopping time to reduce force (Impulse = $F\Delta t$).
- Materials like cardboard absorb energy by deforming over distance.
- Wingsuit provides lift/drag, giving control but still requires high accuracy.
- Targeting is critical — the safety zone is small compared with flight path.